

UNIVERSITI TEKNOLOGI MALAYSIA

Faculty of Computing
SEMESTER 1 SESI 2024 / 2025

LAB ACTIVITY 2

SECP 2523 DATABASE (WBL)

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Let's assume you have a simple database for an **online bookstore** with the following tables:

- **books:** Stores book information.
 - `book_id` (INT)
 - `title` (VARCHAR)
 - `author` (VARCHAR)
 - `genre` (VARCHAR)
 - `price` (DECIMAL)
 - `published_date` (DATE)
- **customers:** Stores customer information.
 - `customer_id` (INT)
 - `first_name` (VARCHAR)
 - `last_name` (VARCHAR)
 - `email` (VARCHAR)
 - `registration_date` (DATE)
- **orders:** Stores order information.
 - `order_id` (INT)
 - `customer_id` (INT) (foreign key from customers)
 - `order_date` (DATE)
- **order_items:** Stores details of books ordered.
 - `order_item_id` (INT)
 - `order_id` (INT) (foreign key from orders)
 - `book_id` (INT) (foreign key from books)
 - `quantity` (INT)

Questions:

Question 1: Retrieve All Books with Their Prices and Published Dates

Write an SQL query to retrieve the title, price, and published_date of all books in the database.

```
SELECT title, price, published_date
FROM books;
```

Question 2: Retrieve Customers Who Registered in the Last 30 Days

Write an SQL query to fetch the first_name, last_name, and email of all customers who registered in the last 30 days.

```
SELECT first_name, last_name, email
FROM customers
WHERE registration_date <= CURDATE() - INTERVAL 30 DAY;
```

Question 3: Find Books by a Specific Author

Write a query to list the title and price of all books written by the author "J.K. Rowling".

```
SELECT title, price
FROM books
WHERE author = 'J.K. Rowling';
```

Question 4: List the Total Number of Orders for Each Customer

Write an SQL query to find the total number of orders placed by each customer. Display their first_name, last_name, and the total_orders.

```
SELECT cust.first_name, cust.last_name, COUNT(o.order_id) AS total_orders
FROM customers cust
JOIN orders o
ON cust.customer_id = o.customer_id
GROUP BY customer_id;
```

**LEFT JOIN or JOIN orders*

Question 5: Find the Most Expensive Book

Write a query to find the title of the most expensive book in the database.

```
SELECT title
FROM books
WHERE price = (SELECT MAX(price) FROM books) ASC
LIMIT 1;
```

**LIMIT 1; to show only one*

**or COUNT BY price DESC LIMIT 1;*

Question 6: Calculate the Total Revenue from All Orders

Write a query to calculate the total revenue generated by the store, considering the quantity of books ordered and their prices.

```
SELECT SUM(b.price*o_quantity) AS total_revenue
FROM order_items o
JOIN books b
ON o.book_id = b.book_id;
```

Question 7: List the Top 5 Customers Who Have Placed the Most Orders

Write a query to retrieve the names of the top 5 customers who have placed the most orders, along with their total number of orders.

```
SELECT cust.first_name, cust.last_name, COUNT(o.order_id) AS total_orders
FROM customers cust
JOIN orders o
ON cust.customer_id = o.customer_id
GROUP BY customer_id
ORDER BY total_orders DESC
LIMIT 5;
```

Question 8: Retrieve Books and Their Total Number of Orders

Write an SQL query to list the titles of books and the total number of times they have been ordered.

```
SELECT b.titles, COUNT(order.item_id) AS total_orders
FROM books b
JOIN order_items order
ON b.book_id = order.book_id
GROUP BY book_id;
```

**GROUP BY book_id; no need b.book_id*

**SUM(order.quantity) = Total Number of Orders*

Question 9: Retrieve Customers Who Have Not Placed Any Orders

Write a query to find all customers who have never placed an order. Display their first_name, last_name, and email.

```
SELECT cust.first_name, cust.last_name, cust.email
FROM customers cust
LEFT JOIN order o
ON cust.customer_id = o.customer_id
WHERE o.order_id IS NULL;
```

**or WHERE customer_id NOT IN (SELECT customer_id FROM orders); no need JOIN*

Question 10: Find the Most Popular Genre Based on Orders

Write an SQL query to find the genre of books that has been ordered the most. Display the genre and the total number of books ordered from that genre.

```
SELECT b.genre, COUNT(o.quantity) AS total_orders
FROM books b
JOIN order_items o
ON b.book_id = o.book_id
GROUP BY b.genre
ORDER BY total_orders DESC
LIMIT 1;
```

**LIMIT 1 to show the most*

Question 11: Find Customers Who Have Spent More than \$500

Write a query to find customers who have spent more than \$500 on their orders. Display the first_name, last_name, and the total amount spent by each customer.

```
SELECT cust.first_name, cust.last_name, SUM(book.price*item.quantity) AS total_spent
FROM customer cust
JOIN orders o
ON cust.customer_id = o.customer_id
JOIN order_items item
ON o.order_id = item.order_id
JOIN books book
ON item.book_id = book.book_id
GROUP BY customer_id
HAVING SUM (book.price*item.quantity) > 500;
```

**or HAVING total_spent > 500;*

Question 12: Find Orders Containing a Specific Book

Write a query to find all orders that contain the book titled "The Catcher in the Rye". Display the order_id and the order_date.

```
SELECT o.order_id, o.order_date
FROM orders o
JOIN order_items item
ON o.order_id = item.order_id
JOIN books book
ON item.book_id = book.book_id
WHERE book.title = 'The Catcher in the Rye';
```

Question 13: Find the Average Price of Books by Genre

Write an SQL query to calculate the average price of books for each genre. Display the genre and the average price.

```
SELECT genre, AVG(price) AS average_price
FROM books
GROUP BY genre;
```

Question 14: Retrieve Orders Placed in the Last Week

Write a query to retrieve the order_id and order_date of all orders placed in the last week.

```
SELECT order_id, order_date
FROM orders
WHERE order_date >= CURDATE() - INTERVAL 7 DAY;
```

Question 15: Retrieve Customers Who Ordered Books by a Specific Author

Write an SQL query to find the first_name, last_name, and email of customers who have ordered books written by the author "George Orwell".

```
SELECT cust.first_name, cust.last_name, cust.email
FROM customers cust
JOIN orders o
ON cust.customer_id = o.customer_id
JOIN order_items item
ON o.order_id = item.order_id
JOIN books b
ON item.book_id = b.book_id
WHERE b.author = 'George Orwell';
```